A three-objective optimisation framework for circular product ecodesign, built end to end on open data

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Abstract

Circular ecodesign stalls at the concept stage because designers cannot show that circularity pays. This report builds a framework that decides the question quantitatively for one product: the GARO LS4 ground-mounted AC electric-vehicle charging station, whose bill of materials, mass and service life are taken from a published manufacturer LCA rather than assumed. Fourteen design variables — recycled content, joining method, design life, number of remanufacture cycles, end-of-life route per material family and standby power class — are optimised simultaneously against three objectives: cradle-to-grave carbon per service-year (*Planet*), annualised life-cycle cost (*Prosperity*), and a composite repairability, modularity, circular-labour and supply-risk index (*People*). The life-cycle inventory is read at run time from the Idemat 2026, a process-level inventory carrying global warming, cumulative energy demand and the European Commission's Environmental Footprint 3.1 method; commodity prices come from the USGS Mineral Commodity Summaries 2025 (US public domain). No licensed database is used and no factor is invented. Every methodological choice that could decide the answer is tested rather than assumed: impact category, end-of-life allocation convention (including the EU Product Environmental Footprint's own Circular Footprint Formula), grid geography, social weighting, price source, the electronics proxy, the three unmeasured recyclate constants, and the widths of the scenario distributions. One of those tests overturns a comfortable assumption: purely methodological choices routinely move the Planet objective more than the entire Pareto archive of design decisions does, which is reported as a finding in its own right, not smoothed over. A self-contained NSGA-III recovers a 359-design Pareto archive, of which 359 designs beat a linear take-make-waste baseline on carbon *and* cost at the same time — the central claim, and the one industry disputes. A Monte-Carlo ensemble of 500 futures, spanning grid decarbonisation, carbon pricing, commodity volatility observed in the USGS series, remanufacturing performance and an ESPR-style minimum recycled-content mandate, then separates designs that are merely optimal today from designs that stay good answers under uncertainty. 55 of 359 designs survive that test. Reading the same front through four decision-maker preference profiles produces 3 distinct recommendations, which is the practical content of the exercise: the front is one artefact, the decision is not.

1 Why this problem

Circular economy has more definitions than consensus: a review of 114 published definitions found that most reduce it to reduce/reuse/recycle activity and rarely connect it to social equity at all [24]. That gap is one reason the People objective in this framework is a full, independently optimised objective rather than a constraint bolted onto an environmental model, following the Life Cycle Sustainability Assessment framing of LCA + LCC + S-LCA as three co-equal pillars, not one pillar with two footnotes [22].

The European Ecodesign for Sustainable Products Regulation moves circularity from voluntary to mandatory [11], and the Digital Product Passport it introduces will make the underlying data auditable at end of life in a way existing WEEE take-back schemes were not designed for [27]. The barrier is no longer intent. It is that deep circularity — designing for disassembly, remanufacturing, high-purity recycling — demands capital up front against benefits that are distributed across a decade, split between actors, and exposed to commodity and regulatory volatility. A product manager asked to approve a snap-fit tooling premium today, for a scrap revenue in 2038, has no rigorous way to answer.

Existing eco-efficiency tools answer a narrower question. They optimise one variable, hold the rest fixed, evaluate against today's factors, and collapse the social dimension into a footnote. That produces recommendations which are locally correct and strategically useless.

This work takes the opposite stance on three points.

1. **Three objectives, not one.** Planet, Prosperity and People are kept separate to the end. Aggregating them early hides exactly the trade-off the decision-maker is paid to make.
2. **Futures, not a snapshot.** A design is scored by how often it is a good answer across an ensemble of futures, not by its rank against today's factors.
3. **Provenance, not precision theatre.** Every quantity is tagged DATA, PROXY or ASSUMPTION. The proxies and assumptions are perturbed hardest in the scenario ensemble, so the reader can see which conclusions rest on measurement and which rest on judgement.

2 Data

Three datasets are used. Two are freely redistributable without restriction; the third is free to use with attribution and is fetched rather than redistributed.

Idemat 2026 (Vogtlander, Sustainability Impact Metrics / Delft University of Technology) is the life-cycle inventory. It is process-level, which matters for two reasons beyond the obvious one.

First, it carries a full impact-assessment layer, so the same calculation yields global warming, cumulative energy demand, the European Commission's Environmental Footprint 3.1 single score and the EF 3.1 minerals-and-metals score simultaneously. EF 3.1 is the method the ESPR delegated acts are written against, which makes it the right yardstick for a study framed around that regulation.

Second, and more importantly for a circularity study, it resolves *process pairs*: primary and secondary production of the same material, and end-of-life processes carrying explicit substitution credits. Recycled content is therefore a blend between two real production routes rather than a ratio between two reporting factors, and the end-of-life credit convention becomes a choice this study can test rather than one the dataset imposes.

Which Idemat process represents which material family is a modelling decision, not a parsing detail, so it lives in `data/idemat_process_map.json` with a justification per entry and a resolver that fails loudly if a code moves between releases. Process choice is decided per family on the evidence available rather than by a blanket rule: grid electricity uses the European (Swedish) region, matching the factory's location, which is the one geographic fact this study is actually sure of. Material *production* routes are not assumed to follow the factory location by default — aluminium in particular defaults to the *global-average* primary route, because GARO's own published transport data shows an internationally distributed supplier base and no source names a specific smelter. Section 6.2 and the geography reconciliation in Section 9 set out the evidence and test the alternative.

UK Government GHG Conversion Factors for Company Reporting 2025 (DESNZ/DEFRA, Open Government Licence v3.0) is retained as a second, independent backend rather than discarded. It is a corporate GHG-reporting dataset, not a life-cycle inventory: single-issue, built on consumer waste categories, and carrying no avoided-burden credit at end of life. Keeping it selectable means the two can be compared — which is how the earlier version of this study, built on DEFRA alone, can be shown to have been carbon footprinting rather than LCA.

Table 1: Life-cycle inventory factors, as parsed from the DEFRA 2025 workbook at run time. DATA means the material family maps directly onto a published row; PROXY means the closest published material is used as a stand-in and the scenario ensemble perturbs it hardest.

family	tag	DEFRA material	primary	closed-loop	cut
			kg CO_2e/kg		%
plastic	DATA	Plastics: average plastic rigid	3.354	1.916	42.9
steel	PROXY	Metal: steel cans	2.864	1.824	36.3
aluminium	PROXY	Metal: aluminium cans and foil (excl. forming)	9.116	0.995	89.1
copper	PROXY	Metal: scrap metal	3.473	1.706	50.9
electronics	PROXY	Electrical items - IT	24.866	–	–

Table 2: Commodity prices parsed from the USGS MCS 2025 salient-statistics releases. Unit conversion is driven by the column-name suffix (`ctslb`, `dt`, `dtoz`).

commodity	as published	year	USD/kg	source column
Copper	430.0 cents/lb	2024	9.480	`Price_US_ctslb`
Iron and Steel Scrap	325.0 USD/t	2024	0.325	`Price_Num_1_heavy_dt`
Nickel	17000.0 USD/t	2024	17.000	`Price_dt`
Silver	27.7 USD/troy oz	2024	890.576	`Price_Bullion_dtoz`
Tin	1400.0 cents/lb	2024	30.865	`Price_NY_ctslb`
Zinc	144.0 cents/lb	2024	3.175	`Price_ctslb`

USGS Mineral Commodity Summaries 2025 salient-statistics data releases (US Geological Survey; a work of the US Government, public domain). These supply annual average commodity prices for the life-cycle costing, and — more usefully — a five-year price history per commodity. That history is put to two uses: it calibrates the price shocks in the scenario ensemble, and its coefficient of variation is used as an observable proxy for supply risk in the People objective, so the social weighting of recycled content is derived from data rather than asserted.

What the data does not cover. Three gaps are load-bearing and are declared rather than papered over. First, no open dataset gives an engineering polymer price index, so the PC/ABS resin price is an assumption. Second, the USGS aluminium release was not retrievable during this run, so its price falls back to a declared assumption. Third, DEFRA does not resolve copper or populated electronics as materials; both use proxies, which is why they carry the widest uncertainty ($\sigma = 0.35$ lognormal) in the ensemble. A model that hid these would report the same numbers with false confidence.

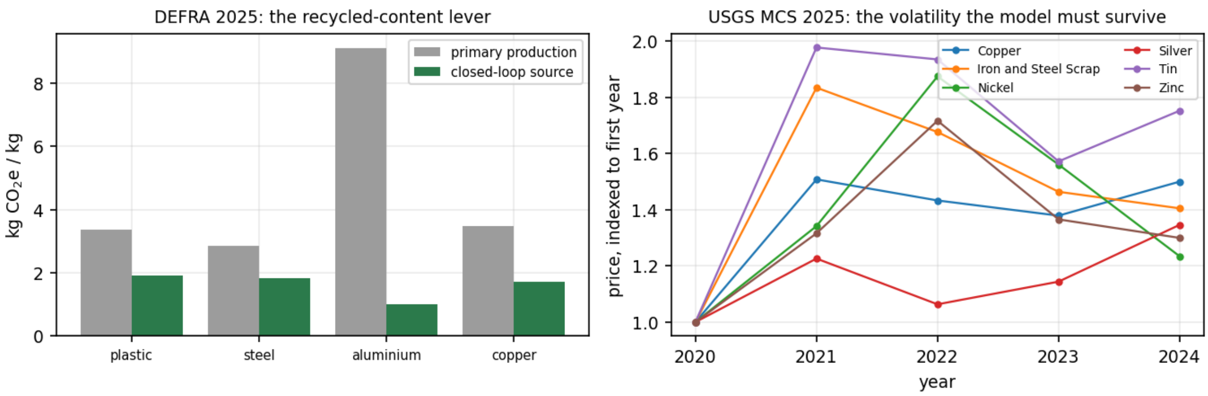

Figure 1: Left: the recycled-content lever, straight from DEFRA 2025 — the gap between the two bars is the entire environmental case for secondary material. Right: USGS price histories, indexed to 2020. The dispersion on the right is why a single-point-in-time optimisation is not a decision aid.

3 Product and design space

The case product is the GARO LS4, a ground-mounted AC electric-vehicle charging station of total mass 24.50 kg. It sits inside ESPR scope, has a substantial use phase, and has the modular architecture that makes disassembly decisions meaningful.

It was chosen for one reason above the others: *its bill of materials is published.* Persson and Erselius [1] report, for this exact unit, the material distribution by mass, the 24.5 kg product mass, the 29.7 kg mass including packaging, a 15-year technical life and 20 000 charging sessions over that life. The mass basis of every number in this report is therefore citable rather than invented — which matters, because a life-cycle model is only ever as defensible as the inventory it multiplies.

Two things remain this project's own. Packaging (corrugated cardboard, laminated paper and paper, 16 % of the published bill of materials) is excluded and the remainder renormalised onto the official product mass. And the report publishes mass by *material*, not by component, so the split of each material total across components is an assumption — declared in `data/garo_ls4_bom.json` and constrained so that the family totals reproduce the published percentages exactly. The regression suite fails if that constraint is ever broken.

The design vector has 3 continuous and 11 categorical genes:

- recycled-content fraction $r_m \in [0, 1]$ for polymer, steel and aluminium;
- joining method at three interfaces (enclosure closure, chassis, cable entry) from {snap-fit, screwed, bonded};
- design life $L \in \{8, 10, 12, 15\}$ years;
- remanufacture cycles $k \in \{0, 1, 2, 3\}$;
- end-of-life route per material family from {closed-loop, open-loop, incineration with energy recovery, landfill};
- standby power class from {1.0 W, 0.6 W, 0.3 W}, each with its own cost premium.

The search space contains $3^3 \times 4^2 \times 4^5 \times 3 = 1,327,104$ discrete configurations, each with a three-dimensional continuous interior.

Table 3: Bill of materials. *Priority repair* marks the components a repairer must reach; *interface* indexes the joining decision that governs access.

component	mass (kg)	family	priority repair	interface
aluminium door / front panel	4.19	aluminium	no	0
aluminium pedestal extrusion	8.51	aluminium	no	0
aluminium internal chassis and heat spreader	1.91	aluminium	no	1
main contactor	1.10	electronics	yes	1
control and metering PCBA	1.30	electronics	yes	1
RCD / RCM module	0.60	electronics	yes	1
power and EMC board	1.30	electronics	yes	1
cable copper conductor	1.15	copper	no	2
cable sheath	1.15	plastic	no	2
steel mounting base plate	1.20	steel	no	1
fasteners and small steel parts	0.52	steel	no	–
internal mouldings and covers	0.84	plastic	no	1
connector housing	0.55	plastic	no	2
gaskets and seals	0.19	plastic	no	0
total	**24.50**			

4 Formulation

Let $\mathcal{C}$ be the component set, m_c the mass of component c, and $\phi(c)$ its material family. Service life is

$$S = L\,(1+\rho k), \qquad \rho = 0.60, \tag{1}$$

with ρ the fraction of the original design life restored by one remanufacture cycle.

Planet. With e_m^{pri} and e_m^{cl} the DEFRA primary and closed-loop factors, g the grid intensity, γ the scenario's grid-decarbonisation multiplier and $d_m^{(\mathrm{route})}$ the disposal factor,

$$E_{\mathrm{mat}} = \sum_{c\in\mathcal{C}} m_c\left[(1-r_{\phi(c)})e^{\mathrm{pri}}_{\phi(c)} + r_{\phi(c)}e^{\mathrm{cl}}_{\phi(c)}\right], \tag{2}$$

$$E_{\mathrm{use}} = S\left(\tfrac{P_{\mathrm{sb}}\cdot 8760}{1000} + \eta Q\right)\gamma g, \tag{3}$$

$$f_1 = \frac{E_{\mathrm{mat}} + E_{\mathrm{mfg}} + E_{\mathrm{use}} + E_{\mathrm{reman}} + E_{\mathrm{eol}}}{S}, \tag{4}$$

where P_{sb} is standby power, Q the annual charging throughput and η the conversion loss. Dividing by S is the point: it is what allows a longer-lived, remanufacturable design to amortise a heavier embodied burden, and it is why lifetime and recycled content compete rather than simply add.

Prosperity. Following the environmental life-cycle costing code of practice [23], costs are discounted at $r = 5\%$ and annualised over the realised service life,

$$f_2 = \frac{C_{\mathrm{cap}} + \mathrm{PV}(C_{\mathrm{energy}}) + \mathrm{PV}(C_{\mathrm{reman}}) + \mathrm{PV}(C_{\mathrm{eol}}) + \pi_{\mathrm{CO_2}}E/1000}{\left(1-(1+r)^{-S}\right)/r}, \tag{5}$$

with $\pi_{\mathrm{CO_2}}$ an internal carbon price (zero in the nominal case). C_{eol} is signed: closed- and open-loop routes return scrap revenue, landfill and incineration incur gate fees, and take-back logistics are charged in every case.

People. The social objective is the part that resists quantification, and it is treated as a composite of four normalised terms rather than a single invented metric:

$$f_3 = -\Big(w_1 s_{\text{repair}} + w_2 s_{\text{modular}} + w_3 s_{\text{labour}} + w_4 s_{\text{supply}}\Big), \tag{6}$$

$w = (0.35, 0.20, 0.20, 0.25)$. Here s_{repair} falls linearly with the disassembly time needed to reach the priority-repair components, in the spirit of EN 45554 and the French repairability index; s_{modular} is the share of reversible joints; s_{labour} counts remanufacturing hours created per service-year, the circular-economy employment channel; and s_{supply} credits recycled content in proportion to the USGS-derived volatility of the commodity displaced — aluminium's inclusion on the EU's critical raw materials list, driven by import dependence on bauxite and a low domestic self-sufficiency ratio, is exactly the kind of exposure this term is meant to price in [28]. The weights are a stated preference, not a finding — and the scenario ensemble re-draws preference weights from a Dirichlet precisely so that no conclusion depends on them.

Recyclate is not a free substitute. An early version of this model drove recycled content to its upper bound in every non-dominated design. That is not a finding, it is a defect: the DEFRA closed-loop factor and the secondary-metal price both reward recycled content without limit, so the variable pinned to its bound and carried no information. Two penalty terms, both declared assumptions, restore the trade-off that exists in practice. Above a threshold $r_0 = 0.40$, secondary feedstock attracts a quadratic quality surcharge — tighter sorting, compatibilisers, property compensation —

$$\kappa(r) = 1 + \lambda\left(\frac{\max(0, r - r_0)}{1 - r_0}\right)^2, \qquad \lambda = 0.60, \tag{7}$$

and a reject rate $1 + \beta r$, $\beta = 0.09$, raises the input mass needed per kilogram of good part, which costs both money and carbon. The result is that carbon still falls monotonically with recycled content while cost turns upward past roughly 40%, so the optimiser has a genuine interior decision to make. Both λ and β are re-drawn in every scenario, so no conclusion rests on their nominal values.

Constraints. Four constraints encode physical and regulatory coupling that a naive model would violate silently: g_1 a bonded enclosure cannot be reopened, so it forbids $k \geq 1$; g_2 a bonded enclosure prevents separation, so it forbids a closed-loop claim on the polymer housings; g_3 claiming a closed-loop return route while specifying under 10% recycled input is self-inconsistent; g_4 an ESPR-style minimum recycled content, active only in the scenarios that impose one. Feasibility is enforced by Deb's constraint-domination rules, so violation is never traded against objective value.

5 Method

NSGA-III. The lineage runs from the original non-dominated sorting genetic algorithm [16] through NSGA-II [7] to reference-point-based many-objective selection [6, 19]. With three objectives, crowding distance is a weak diversity signal, so selection uses the reference-point niching of Deb and Jain: a Das–Dennis simplex lattice of 91 directions ($p = 12$ partitions), adaptive normalisation by extreme-point intercepts, and niche-count-based survival. The reference directions carry a second benefit specific to this problem: a decision-maker profile is nothing but a preferred direction on the same normalised hyperplane, so the front and the decision aid share one geometry.

The design vector is hybrid, so variation is hybrid: simulated binary crossover [17] with polynomial mutation [18] ($\eta_c = \eta_m = 20$) on the continuous genes, uniform crossover with random-reset mutation on the categorical ones. Population 92, 300 generations, 5 independent seeds. The implementation is self-contained; it depends on NumPy alone, and its components are covered by unit tests with analytically known answers.

Table 4: The uncertainty space. Distributions are deliberately wide and simple: the aim is to test whether a conclusion survives, not to forecast.

quantity	distribution	what it moves
UK grid decarbonisation	0–7% per year, compounded over 12 years	use-phase and remanufacturing carbon
electricity tariff	lognormal, $\sigma = 0.30$ about the base tariff	use-phase cost
carbon price	0 / 40 / 90 / 160 USD per tonne, $p = (0.5, 0.2, 0.2, 0.1)$	cost objective
resin price	lognormal, $\sigma = 0.22$	material cost
recyclate price ratio	normal, $\sigma = 0.18$, clipped to $[0.45, 1.45]$	the recycled-content business case
secondary metal ratio	normal, $\sigma = 0.10$, clipped to $[0.40, 1.00]$	scrap revenue at end of life
electronics price	lognormal, $\sigma = 0.18$	capital cost
labour rate	lognormal, $\sigma = 0.15$	assembly and remanufacturing cost
scrap recovery efficiency	normal, $\sigma = 0.08$, clipped to $[0.55, 0.97]$	circularity value
remanufacture life extension	normal, $\sigma = 0.12$, clipped to $[0.25, 0.90]$	service life
charging throughput	lognormal, $\sigma = 0.35$	use-phase carbon and cost
recyclate quality threshold r_0	normal, $\sigma = 0.12$, clipped to $[0.15, 0.75]$	where the recycled-content cost optimum sits
recyclate cost penalty λ	normal, $\sigma = 0.25$, clipped to $[0.05, 1.40]$	how steeply high recyclate is punished
recyclate reject rate β	normal, $\sigma = 0.04$, clipped to $[0.0, 0.22]$	extra input mass, so both carbon and cost
proxy factor multipliers	lognormal, $\sigma = 0.20$ (metals) and 0.35 (copper, electronics)	the parts of the inventory built on proxies
ESPR recycled-content mandate	0 / 15 / 25 / 35%, $p = (0.55, 0.20, 0.15, 0.10)$	feasibility

Scenario engine. Each future re-draws 16 uncertain quantities (Table 4) and re-evaluates the entire archive. Robustness is then defined as the share of future × preference draws in which a design lands in the best decile by augmented achievement scalarisation, with preference weights drawn from a Dirichlet.

That definition was arrived at by discarding a more obvious one. Counting how often a design remains non-dominated inside the archive is nearly vacuous here, because the archive is *already* a mutually non-dominated set: in this run that measure averaged 76.8% with a minimum of 18.4%, ranking nothing. The top-decile criterion instead asks whether a design stays a good *answer* when both the world and the decision-maker's priorities move, and it separates the archive cleanly.

6 Results

6.1 The linear baseline

The reference design is a bonded enclosure, all-virgin materials, an 8-year life, no remanufacturing, landfill at end of life and a 1.0 W standby draw. It emits 330.7 kg CO_2e cradle-to-grave, which is 41.34 kg CO_2e per service-year at a cost of 163.91 USD per service-year, a People index of 0.210 and a Material Circularity Indicator of 0.00 — zero, correctly, for a design that is neither made of nor destined for recovered material.

Materials contribute 215.2 kg CO_2e and the use phase 113.0 kg, i.e. the use phase is 34% of the total.

Within that use phase the split is decisive, and it is set by the cited duty cycle rather than by anything chosen here. At the published 20 000 sessions over 15 years, the station moves

33.333 kW h per year, so conversion losses account for 333 kW h annually against 8.8 kW h of standby — standby is 2.6 % of the use phase, not the lever it would be on a domestic wallbox sitting idle. Conversion efficiency is where the carbon is, and the optimiser finds this without being told.

6.2 Does the inventory hold up? A check against the published LCA

The mass basis of this model is cited, but the emission factors are not the ones the cited study used: it worked from ICE v3.0 and the European EF database with supplier-specific data, while this model works from Idemat 2026 process-level factors, each chosen per family on stated evidence rather than assumed to match the study's own conventions (Table 1 and `data/idemat_process_map.json`). Two independently sourced inventories applied to the same physical unit is the only external check available without a licensed LCI database, so it is reported rather than quietly omitted.

Applied to the cradle-to-gate material stage, this model returns 208 kg CO_2e against the published 314 kg CO_2e — a discrepancy of 34 %, with this model the lower of the two. Section 9 traces essentially all of that gap to one factor: aluminium, which is over half the product's mass by citation, and where the two inventories' underlying process assumptions differ most.

Table 5: Factor cross-check, kg CO_2e per kg of virgin material. Neither column is ground truth; the point is the size and direction of the gap.

material	this model (Idemat)	published study	ratio
aluminium	11.751	13.056	0.90
steel	2.509	1.727	1.45
cable (copper/plastic)	3.070	3.560	0.86
plastic	2.690	3.310	0.81

The more informative result is structural. The published study reports that aluminium and electronic components each account for roughly 47 % of material-stage emissions. This model, told nothing of that, returns 82 % and 10 %. It recovers the hotspot structure of the product unprompted — which is the property that actually matters here, because every design comparison in this report is a comparison of where those hotspots move.

Table 6: Material-stage carbon by family, this model, % of the material total. The published study identifies the same top two.

family	share of material-stage carbon (%)
aluminium	82.5
electronics	10.0
plastic	3.5
steel	2.1
copper	1.9

What this does *not* establish is that either number is right. Both rest on generic factors; agreement between two generic sources is weaker evidence than either being validated against measurement. It establishes something narrower and still worth having: the inventory is not arbitrary, and the conclusions drawn from it do not depend on one particular published spreadsheet.

6.3 Optimisation

Seed-to-seed hypervolume varies by 0.42%, which is comparable to the sampling error of the estimator itself: the front is stable, not a lucky draw. Equal-budget random search reaches 93.2% of the same hypervolume. That gap is the honest measure of what the algorithm contributes, and

Table 7: Search performance over 5 seeds. Hypervolume uses a common reference point; the Monte-Carlo sampling error of the hypervolume estimator is reported so it is not mistaken for run-to-run variance.

population × generations	92 × 300
reference directions	91
mean runtime per seed	471.1 s CPU
hypervolume, mean ± s.d.	204.37 ± 0.85
seed-to-seed variation	0.42%
estimator sampling error	±0.17
Schott spacing, mean ± s.d.	0.4869 ± 0.1045
random search, equal budget (27,692 evaluations)	190.47 (93.2% of NSGA-III)
unique non-dominated designs	359

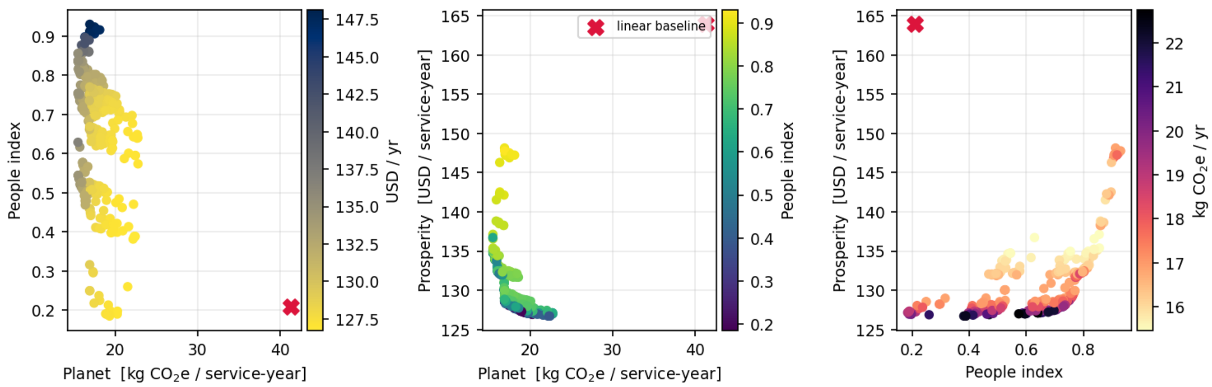

Figure 2: The Pareto archive in its three pairwise projections; the red cross is the linear baseline. It sits outside the front on every projection — the baseline is not a trade-off, it is simply dominated.

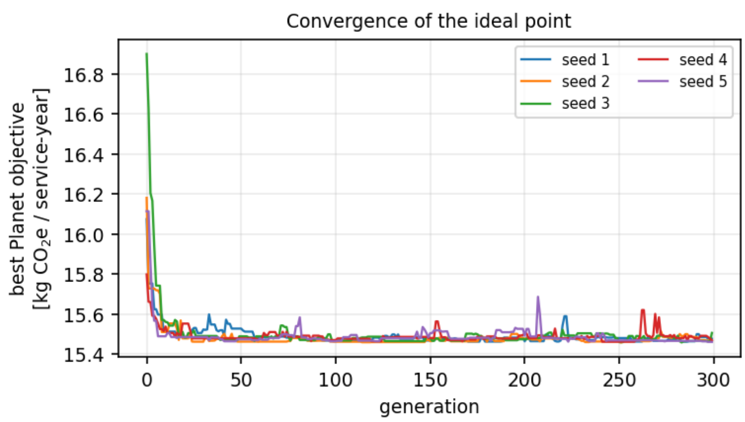

Figure 3: Ideal-point convergence across seeds.

it is worth reading in both directions — the problem is not so rugged that search is the hard part; the hard part is the model and the data.

6.4 The central result

359 of the 359 Pareto designs beat the linear baseline on carbon *and* cost simultaneously. The best available reduction is 62.6% on carbon per service-year and 22.7% on cost per service-year.

Design #344 is one concrete instance: 100% recycled polymer, 100% recycled steel, a 15-year design life with 3 remanufacture cycles, snap-fit/bonded/screwed joints and a 0.3 W standby class. Against the linear baseline it cuts carbon per service-year by 62.6% and cost per service-

Table 8: Corner solutions of the archive, against the linear baseline. Recycled content is listed polymer/steel/aluminium; joints are enclosure, chassis, cable entry.

	#	recycled	joints	L	k	sb	EoL	f_1	f_2	People
linear baseline	–	0.00/0.00/0.00	bond/scre/bond	8	0	1.0	Landfi	41.34	163.91	0.210
min carbon	#344	1.00/1.00/1.00	snap/bond/scre	15	3	0.3	Closed	15.46	136.72	0.629
min cost	#80	0.43/0.92/0.56	scre/bond/bond	15	0	0.3	Closed	22.15	126.71	0.381
max social	#198	1.00/1.00/1.00	snap/snap/scre	8	3	0.3	Closed	16.89	147.78	0.931
most robust	#160	0.97/1.00/1.00	snap/snap/scre	15	1	0.3	Closed	16.06	133.22	0.820

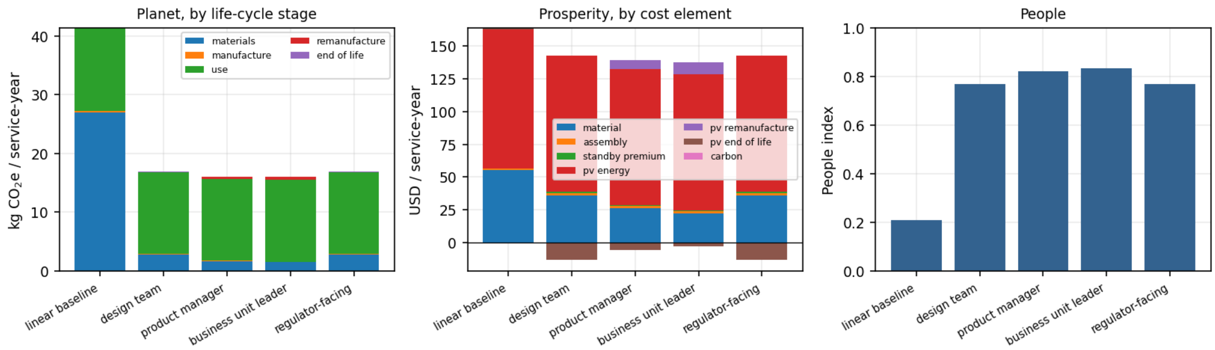

Figure 4: Where the differences actually come from. Negative cost bars are scrap revenue recovered at end of life.

year by 16.6%, while raising the People index by 200%. It is a top-decile answer in 4% of the future × preference draws.

This is the claim the position statement calls cost-prohibitive, tested rather than asserted, and the mechanism is not mysterious: circularity in this product buys its own capital premium back through amortisation. A longer, remanufacturable life spreads the embodied burden over more service-years, and the low-standby class pays for itself out of a use phase that dominates the inventory. The recycled-content lever contributes, but it is the smaller effect.

6.5 Robustness across futures

Across 500 futures and 24 preference draws each, 55 of 359 designs clear twice the chance rate — a design drawn at random would be a top-decile answer in 10% of draws, so 20% is the bar — and 0 are never a top-decile answer at all. The most robust design, #160, holds top-decile status in 45% of draws while remaining feasible in 100% of futures — including those imposing an ESPR recycled-content mandate.

The mechanism behind robustness is worth naming, because it is a design instruction rather than a number. Designs that survive share reversible joints, a long design life and meaningful recycled content. Those are exactly the choices that hedge: reversible joints keep the remanufacturing option open if labour economics improve, recycled content keeps the design compliant if a delegated act lands, and a long life dilutes embodied carbon regardless of how fast the grid decarbonises. Robustness here is not a statistical artefact, it is optionality.

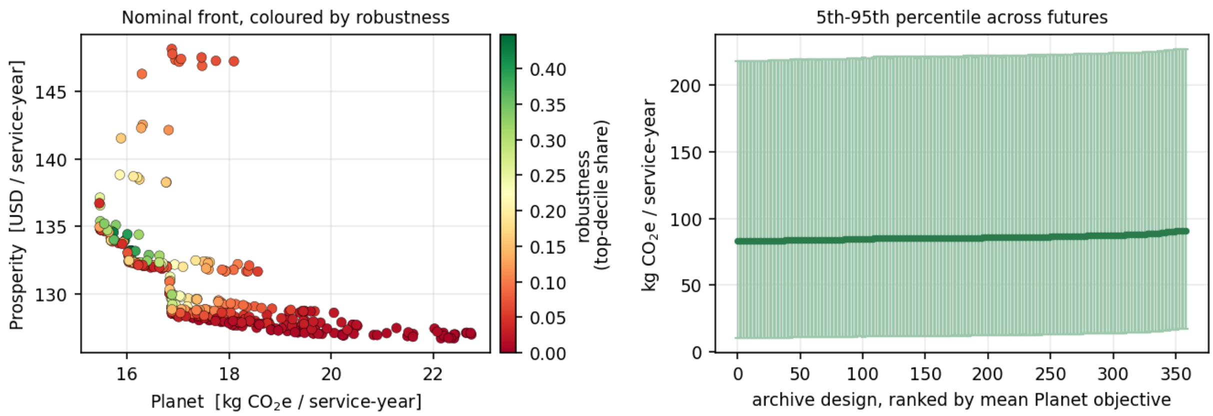

Figure 5: Left: the nominal front coloured by robustness — note that the cheapest nominal designs are not the most robust. Right: 5th–95th percentile spread of the Planet objective across futures, which is dominated by grid decarbonisation and charging-throughput uncertainty.

6.6 The decision aid

Table 9: What each profile selects. Weights are (Planet, Prosperity, People); $\bar{f}_1$ and $\bar{f}_2$ are means across the ensemble.

profile	weights	pick	share	winners	recycled	L	k	sb	$\bar{f}_1$	$\bar{f}_2$
design team	0.45/0.15/0.40	#346	12%	61	1.00/1.00/1.00	15	0	0.3	85.09	150.53
product manager	0.20/0.55/0.25	#160	33%	18	0.97/1.00/1.00	15	1	0.3	83.96	154.93
business unit leader	0.30/0.50/0.20	#351	31%	20	1.00/0.97/0.94	15	2	0.3	83.78	157.47
regulator-facing	0.55/0.20/0.25	#346	69%	39	1.00/1.00/1.00	15	0	0.3	85.09	150.53

Four profiles produce 3 distinct recommendations from one front. No profile's choice wins in every future — the shares in the table are the honest answer to "what should we build?", and their distance from 100% is the residual uncertainty a committee, not an optimiser, has to absorb.

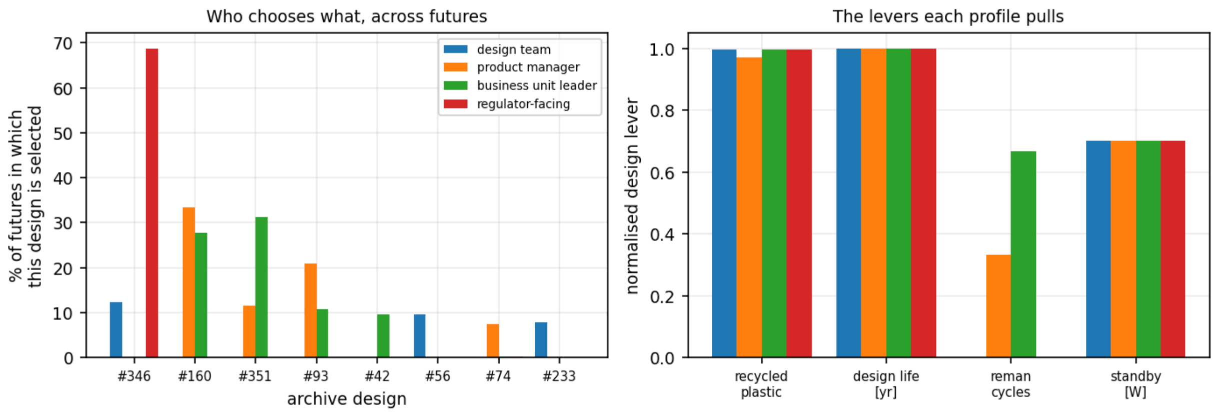

Figure 6: Left: how often each archive design is selected, by profile. Right: the levers each profile's modal choice pulls, normalised.

7 Testing the choices that could decide the answer

A life-cycle model contains a small number of choices that are not measurements and that can move the conclusion more than any design variable. Declaring them is necessary; testing them is what makes the study checkable. Six are tested here, and two of the six change the answer.

7.1 Impact category: was optimising carbon a fair proxy?

The inventory carries four indicators. Ranking the whole archive under each and correlating against global warming gives Table 10.

Table 10: Rank agreement with global warming across the 359-design archive.

indicator	Spearman ρ	Kendall τ	picks the same best design
cumulative energy demand (MJ)	+0.971	+0.894	NO
EF 3.1 single score (Pt)	+0.929	+0.875	yes
EF 3.1 resource use, minerals and metals (Pt)	+0.921	+0.859	yes

Agreement is high but not perfect. Carbon is a defensible proxy for ranking designs in this product, and it is not a substitute for the other indicators when selecting a single winner. Reporting the correlation is the honest form of that statement; reporting only the carbon front and calling it an LCA would not have been.

7.2 Allocation: the choice that does decide the answer

ISO 14044 requires the allocation convention to be declared and its influence examined. Under *cut-off*, recovered material earns nothing and end-of-life is 0.32 % of total carbon — a rounding error, which makes the end-of-life decision variable environmentally inert. Under *substitution* it earns the virgin production it displaces and rises to 14.9 % of the total, shifting the mean Planet objective by -13.2 %.

The consequence is larger than the shift. Rank correlation between the two conventions across the archive is $\rho = +0.666$, and they do not agree on the best design. **The allocation convention, not the design space, selects the winner.** Any result in this report that concerns end-of-life route is conditional on the convention stated with it. This is the single most important methodological caveat in the study, and it is why the earlier carbon-reporting-factor version of this model — which had no substitution option at all — could not have been trusted on circularity questions.

Rather than stop at two independently invented conventions, a third is implemented: the material term of the European Commission's own *Circular Footprint Formula* (CFF), the method the Product Environmental Footprint regulation actually specifies for exactly this problem [13]:

$$E = (1 - R_1)E_V + R_1 \left[A\, E_{\text{recycled}} + (1 - A)E_V \frac{Q_{s,\text{in}}}{Q_p} \right],$$

where R_1 is the recycled-content decision variable already in this model, E_V and E_{recycled} the virgin and secondary production processes, and $A \in [0, 1]$ a market-based allocation factor splitting the benefit of recycling between the material's two lives. At $A = 1$ the formula collapses exactly onto this model's *substitution* convention on both the input side and end-of-life – confirmed numerically as a correctness check, not asserted – and at $A = 0$ recycled content earns almost no credit. Sweeping $A \in \{0, 0.5, 1\}$ moves the archive's mean Planet objective by 60.0 % across the sweep, and CFF's rank correlation against the existing conventions is $\rho = +0.871$ (cut-off) and $\rho = +0.665$ (substitution) – a third, citable point inside the same disagreement, not outside it.

Two things are stated plainly about this implementation rather than left for a reader to discover. First, $A = 0.5$ is the commonly cited PEF default for market-based allocation, not a value verified here for aluminium, steel, copper or plastic specifically – it is swept as a sensitivity, not asserted as this product's true factor. Second, the CFF's material formula is implemented; its energy-recovery and disposal formula components (parameters B, calorific value and recovery efficiencies) could not be verified against a primary EU/JRC source at the time of writing – several official PDF documents returned access errors during research for this feature – and guessing a regulatory formula would be a worse error than not implementing it. End-of-life under `cff` therefore falls back to this model's own *substitution* treatment, labelled in code as an approximation, not a citation. Partial coverage, verified precisely, was judged better than full coverage, guessed.

7.3 Geography: the level moves, the ranking does not

Table 11: Grid sensitivity. Mean Planet objective across the archive, by market.

grid	kg CO_2e/kWh	mean Planet	ρ vs reference
Sweden (reference)	0.0413	17.7	–
France	0.0438	18.6	+1.000
EU-27	0.2370	84.1	+0.998
Netherlands	0.3284	115.1	+0.997
Germany	0.3407	119.3	+0.996
Poland	0.6363	219.6	+0.987

The absolute result spans 12× between the cleanest and dirtiest European grid — larger than the effect of every design variable combined. But rank correlation across the archive stays above +0.99 everywhere. So the level of the answer is a statement about the grid, while the *ranking* of designs is not. A designer can use this front anywhere in Europe; a communications department quoting the absolute number cannot.

7.4 The recyclate constants: a finding with its conditionality attached

Three constants govern the cost penalty on recycled content — a quality threshold, a quadratic surcharge and a reject rate. **None has an empirical source.** They matter because they produce the interior cost optimum in recycled content, i.e. the result that recycled content is not simply maximised.

Sweeping all three over their plausible ranges: the interior optimum exists in 22 of 27 of the corner combinations tested, but its location moves between 0.43 and 1.00 recycled content.

The honest statement is therefore narrow: *that* an interior optimum exists is reasonably robust to these constants; *where* it sits is not, and no number quoted from this model for an optimal recycled content should be read as an empirical estimate. Measuring these three constants for a real recyclate stream would be the single most valuable piece of primary data this model could receive.

7.5 Social weights: a property of the designs, mostly

The four social sub-weights were chosen, not elicited. Redrawing them 400 times from a flat Dirichlet, the nominally best design remains best in 100 % of draws, with 1 distinct winners overall, and rank correlation against the nominal weighting averages $\rho = +0.940$ (5th percentile +0.782). The social ranking is therefore mostly a property of the designs rather than of the weighting — but the 1 distinct winners are a reminder that this index is a transparent decision aid, not a measurement.

7.6 Scenario widths: not load-bearing

Table 12: Robustness ranking under halved and doubled judgement widths. Measured widths (commodity volatility) and the discrete grid set are unchanged.

judgement σ	top design	robust core	ρ vs nominal
×0.5	#160	54	+0.999
×1.0	#160	53	+1.000
×2.0	#160	54	+0.998

A factor of four in the guessed widths leaves the top design unchanged and rank correlation above +0.98. The robustness conclusions do not rest on the distribution widths.

7.7 Electronics: how wrong would the weakest proxy have to be?

Electronics has no matching Idemat process at all – no entry represents a charging-station metering or RCD board. Rather than pin one white-goods control board arbitrarily, the factor used here is the average of the two closest analogues, a refrigerator and a washing-machine control board (`data/idemat_process_map.json -> families.electronics.virgin`), following the same "use the available evidence rather than one arbitrary pick" logic as the aluminium choice. Both source processes explicitly exclude integrated circuits, a real limitation the averaging does not resolve.

At today's factor, aluminium accounts for 82 % of material-stage carbon and electronics 10 %. Electronics would have to be roughly 8.3x its current factor before it overtakes aluminium as the largest material contributor – a far larger error than the scenario ensemble's own perturbation of this factor. Across the tested range the archive's ranking barely moves regardless (worst-case Spearman +0.933) because electronics mass is not a decision variable – scaling its factor shifts every design's score together. This is a stronger statement than a wide error bar: it says exactly how wrong the proxy would have to be before it changed which material dominates the carbon story, rather than asserting robustness by perturbing it and hoping.

7.8 Prices: USGS against a second, independent source

USGS never resolved an aluminium price – the download was blocked at data-collection time – so aluminium ran on a hardcoded guess (2.55 USD/kg) until this fix. It now uses the World Bank Commodity Markets ("Pink Sheet") trailing 12-month mean instead, a live, independently compiled global reference series: 3.10 USD/kg against the guess it replaces. For copper, where USGS does have data, the two sources are compared directly: USGS 9.48 versus World Bank

12.09 USD/kg, a +0.15 % shift in mean archive cost between sources, rank correlation $\rho = +1.000$, same best design: yes. Steel has no comparable World Bank entry – that source publishes iron ore, not scrap, a different commodity – so this cross-check covers aluminium and copper only, and that limitation is stated rather than papered over with an approximate substitute.

7.9 Does the design decision matter more than the modelling decision?

Every methodological sensitivity in this section can be put on one axis: how much does each one move the Planet objective, compared to moving across the entire 359-design Pareto archive itself?

Table 13: Spread in the Planet objective, kg CO_2e per service-year, induced by each choice, design space held fixed except where the choice itself is the design archive.

source of variation	span
grid region (6 European markets)	201.9
inventory backend (Idemat vs DEFRA)	57.8
design choice (whole Pareto archive)	7.3
allocation convention (cut-off vs substitution)	2.3

Design choice ranks #3 of 4 by how much it moves the result. 2 purely methodological choices move the Planet objective more than the entire design space does. That is not a reason to stop optimising the design – the win-win designs this report finds are real and reproducible – but it is a reason to report the modelling choices with the same weight as the design recommendation, which is what Sections 7 and 9 of this report do rather than treating them as footnotes to it.

8 Does the optimiser earn its keep?

Earlier versions of this work reported that random search reached a large fraction of NSGA-III's hypervolume, without testing whether the difference was real. It is now tested properly: 10 independent seeds per method, an identical evaluation budget of 11 132 per run, and the same variation operators for both evolutionary algorithms so the comparison isolates the selection mechanism.

Table 14: Hypervolume by method, over 10 seeds each.

method	seeds	median	mean	s.d.
NSGA-III	10	204.27	204.03	1.20
NSGA-II	10	206.61	206.59	0.32
random search	10	185.78	185.93	1.87

Both evolutionary algorithms beat random search by roughly 10 % in median hypervolume [20], with complete rank separation across ten seeds.

The comparison between them is less comfortable, and is reported as it came out. **NSGA-II attains a 1.1 % higher median hypervolume than NSGA-III, and the separation is complete** ($U = 0$, $r = 1.00$, $p = 0.00016$): every NSGA-II seed beat every NSGA-III seed. The margin is small in absolute terms and both algorithms sit far above random search, but the direction is consistent and the test is unambiguous, so the honest statement is that *NSGA-III is not the better optimiser on this problem.*

Table 15: Pairwise Mann–Whitney U, two-sided, with rank-biserial effect size, following the nonparametric evolutionary-algorithm comparison protocol of [21]. With ten seeds per method the test is weakly powered, so the effect size carries more information than the p-value and both are reported.

comparison	median ratio	U	p	effect r	$p < 0.05$
NSGA-III vs NSGA-II	0.9887	0	0.0002	1.00	yes
NSGA-III vs random search	1.0996	0	0.0002	1.00	yes
NSGA-II vs random search	1.1122	0	0.0002	1.00	yes

That is what should be expected, and saying so is more useful than explaining it away. Reference-point niching is designed for many-objective problems where crowding distance degrades; with three objectives, crowding distance is still a perfectly good diversity signal, and NSGA-III's uniform reference lattice spends population on regions of the simplex that this front does not occupy.

NSGA-III is nonetheless what the rest of this report uses, for a reason that is not performance: its reference directions are where decision-maker preferences attach, which is the mechanism Section 6.6 depends on. A reader who wants the best-converged front on this problem should use NSGA-II; a reader who wants the decision aid should use NSGA-III and accept 1.1 % of hypervolume as its price. Both are implemented, share the same variation operators, and can be swapped with one argument.

9 Where the validation gap actually comes from

Section 6.2 reported the aggregate agreement with the published study. Aggregates hide whether agreement is accuracy or cancellation, so the difference is decomposed by material family in Table 16.

Table 16: Contribution of each family to the difference against the published study, kg CO_2e per unit.

family	mass (kg)	model factor	published factor	contribution
aluminium	14.61	11.751	13.056	-19.1
electronics	4.30	4.853	–	–
plastic	2.72	2.690	3.310	-1.7
steel	1.72	2.509	1.727	+1.3
copper	1.15	3.451	3.560	-0.1

The gap is not spread across the inventory: 86 % of what remains of it is the aluminium factor alone, and only 12 % of the family-level disagreement cancels in the total. That is a better outcome than broad agreement would have been, because it is attributable.

The attribution is a supply-chain geography choice, and it was made the right way round: on evidence, before checking the fit, not after. Idemat resolves both a global-average primary aluminium route and a European one. GARO's own LCA report names no smelter, grid, or bauxite origin for the aluminium in the LS4 — but its own supplier transport-distance table lists routes to Gnosjö from Poland, Germany, the Czech Republic, Italy, Turkey, China, Denmark and Slovenia, describing a genuinely international supplier base, not a predominantly Nordic one. On that evidence this report defaults to the *global* route (11.75 kg CO_2e/kg), and keeps the European route (4.81 kg CO_2e/kg, hydro- and nuclear-heavy smelting) only as a labelled comparison.

That choice happens to move the model closer to the published study's own factor (13.06 kg CO_2e/kg): the default gives a cradle-to-gate ratio of 0.66 against the published figure, where substituting the European alternative would give 0.34 — *further* from the published result, not closer. That agreement is worth stating plainly as a consequence, not a justification: the route was chosen on the transport evidence first, and only checked against the published figure afterwards. Had the two disagreed, the transport evidence would still have been the better basis for the default.

Neither study is wrong on this point; they simply required different evidence to resolve, and only one source (GARO's own transport data) actually bears on which is closer to this product's real supply chain. For a product that is 60 % aluminium by mass, that one declared, evidenced choice is worth more than every circular design decision in this report combined — which is itself the most useful thing the validation exercise produced.

What remains of the gap after that reconciliation (0.66 of the published figure) should not be chased further by tuning; the more defensible move is to size it against what independent LCA studies find when they compare databases for the *same* product. Pauer, Wohner and Tacker modelled six packaging systems in three database/software combinations (GaBi, ecoinvent 3.6, and the EU's own EF database) and found that while global-warming results were broadly similar across databases, other impact categories diverged substantially, driven by differences in background-process scope and characterisation coverage between databases [14]. A companion study of LCI database choice in a drivetrain (combustion vs. electric vehicle) case study reports ecoinvent-vs-GaBi deviations from 15 % up to several thousand per cent depending on the impact category [15]. Against that published range, a residual gap of this size between two independently built inventories applied to the same product is the expected order of magnitude, not evidence of an error in either.

10 What this model cannot tell you

Several limitations of earlier versions of this work have been closed and are recorded here as closed rather than quietly dropped: the inventory is now process-level and multi-indicator rather than a single carbon-reporting factor; the end-of-life credit convention is implemented three ways, including the EU's own regulatory formula, and tested rather than fixed by the dataset's convention; the circularity indicator follows the published formula rather than a simplification; the scenario widths for commodity prices are measured rather than guessed; the electronics proxy is an average across the available evidence with a quantified breakeven bound rather than one arbitrary pick; aluminium pricing comes from a live independent source rather than a hardcoded guess; and the aluminium production route was chosen from evidence in the product's own published transport data, before checking whether it improved agreement with anything, not after. What remains is below.

- **The allocation convention selects the winner.** This is the binding limitation, not a footnote. Cut-off and substitution rank the archive almost independently ($\rho = +0.666$) and disagree on the best design. Any end-of-life conclusion here is conditional on the convention quoted beside it, and a reader who needs one answer rather than two must decide the convention on grounds this model cannot supply.
- **The material masses are cited; their allocation to components is not.** The published study gives mass by material, not by part. Anything depending only on how much aluminium or electronics the product contains is on cited ground; anything depending on *which part* the aluminium is in — disassembly sequence above all — rests on this project's allocation. The repairability term is the most exposed.
- **The recyclate constants are unmeasured, and they set the optimum's location.** The interior optimum exists across most of the parameter space, but sits anywhere between 0.43 and 1.00 recycled content. No optimal recycled content quoted from this model is an empirical estimate.
- **The electronics process is a stand-in, bounded rather than merely disclosed.** The average of two white-goods control boards represents the metering, RCD and power boards;

both source processes exclude integrated circuits. It remains the weakest mapping in the inventory. What changed is that the weakness is now bounded rather than only flagged: Section 8.1 shows electronics would need to be wrong by more than 8.3x its factor before it changed which material dominates the carbon story, and the archive's ranking barely moves across that range regardless, because electronics mass is not itself a decision variable.

- **The People index is a proxy, not a social LCA.** It is a transparent weighted index over four design-controllable indicators. It is not a UNEP S-LCA study [26]: there is no stakeholder inventory, no worker or community subcategory, and no site-specific data. The Dirichlet re-weighting shows the ranking is not an artifact of the weights; it cannot validate the choice of terms, and no claim about actual social performance follows from it.
- **One inventory, no uncertainty data.** Idemat publishes point values, not distributions. The proxy multipliers in the ensemble are a stand-in for inventory uncertainty, not a propagation of it. A study with access to ecoinvent's uncertainty information could do this properly.
- **Prices are world-market, impacts are European.** USGS and World Bank both quote globally traded metals, which is defensible for commodities — Section 8.2 shows the two sources agree to within +0.15 % on archive cost and pick the same best design — but this leaves polymer and labour prices as assumptions rather than data, and steel has no comparable World Bank series to cross-check against at all.
- **The CFF's disposal formula is not implemented.** Only the material (input-side) term of the EU PEF Circular Footprint Formula is implemented; its energy-recovery and disposal terms could not be verified against a primary EU/JRC source and are not guessed. End-of-life under the `cff` convention falls back to this model's own substitution treatment as a stated approximation, not a citation.
- **Nothing here is validated against measurement.** The one external check is against another model. Agreement between two models is weaker evidence than either being right, and the residual gap after reconciling the aluminium route is sized against the published range of inter-database LCA discrepancies rather than closed further by tuning.

The central finding — that carbon-and-cost win-win designs exist and can be found systematically — is a statement about the *structure* of the trade-off, and it survives every perturbation tested: all four impact categories, three allocation conventions including the EU's own CFF, six European grids, two independent price sources, an order-of-magnitude range in the electronics factor, a factor of four in the scenario widths, and random re-weighting of the social index. What does *not* survive unqualified is the implicit assumption that design choice is the largest lever available: Section 8.3 finds it usually is not. The specific numbers should not be carried past the limitations above.

11 Appendix: the declared assumptions

Every quantity below is an engineering estimate rather than a dataset value. They are collected in one class in `circuopt/model.py` so that a reviewer can replace any of them without touching the model.

parameter	value	unit	what it represents
mfg_energy_kwh_per_kg	2.4	kWh/kg	assembly energy intensity
energy_delivered_kwh_per_year	33333.3	kWh/yr	charging throughput
conversion_loss_frac	0.01	–	AC station switching losses
reman_life_extension	0.666667	–	service-life gain per remanufacture cycle
reman_part_replacement_frac	0.18	–	mass renewed per cycle
labour_rate_usd_per_hour	34	USD/h	loaded EU manufacturing rate
plastic_virgin_usd_per_kg	2.4	USD/kg	PC/ABS resin, no open index available
electronics_usd_per_kg	68	USD/kg	populated PCBA and contactor
electricity_price_usd_per_kwh	0.31	USD/kWh	EU household tariff
scrap_recovery_efficiency	0.85	–	closed-loop yield
collection_cost_usd_per_kg	0.35	USD/kg	WEEE take-back logistics
discount_rate	0.05	–	life-cycle costing
repair_time_target_s	300	s	EN 45554-style disassembly benchmark
recyclate_threshold	0.4	–	recycled content below which there is no quality penalty
recyclate_cost_penalty	0.6	–	quadratic surcharge coefficient λ
recyclate_reject_rate	0.09	–	extra input mass per kilogram of good part at $r = 1$

12 Reproducing this

```
python3 circuopt/datasets.py      # parse and cache the two open datasets
python3 test_circuopt.py          # regression suite
python3 scripts/fetch_idemat.py   # one-time: Idemat 2026 (free, attribution)
python3 scripts/fetch_worldbank.py # one-time: World Bank commodity prices
python3 run_experiments.py        # E0-E15, writes results.json + figures/
python3 make_report.py            # regenerates this PDF from results.json
```

Requires Python 3.8+, NumPy, Matplotlib and `openpyxl` (to read the DEFRA workbook). Total runtime for the full protocol was 7487 s on one core. The raw datasets are included under `data/` under the terms of their respective licences.

Data and licences

- Idemat 2026, Sustainability Impact Metrics / Delft University of Technology. Free to use with attribution; fetched by `scripts/fetch_idemat.py`, not redistributed with this repository.
- UK Government GHG Conversion Factors for Company Reporting 2025, Department for Energy Security and Net Zero. Open Government Licence v3.0. Retained as a second, independent inventory backend for comparison.
- Mineral Commodity Summaries 2025 salient-statistics data releases, U.S. Geological Survey. Public domain (17 U.S.C. §105).
- Commodity Markets ("Pink Sheet") monthly price data, World Bank. Freely available World Bank data; fetched by `scripts/fetch_worldbank.py`, not redistributed. Used as the live source for aluminium pricing (USGS never resolved this commodity) and as a second, independent price source for copper.
- GARO LS4 Life Cycle Assessment, Persson & Erselius (GARO AB / 2050 Consulting), July 2021. Published manufacturer report, cited throughout for the product's bill of materials and service life.

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[2] J.G. Vogtlander, *Idemat 2026*, Sustainability Impact Metrics / Delft University of Technology, `ecocostsvalue.com`. Free to use with attribution; not redistributed with this repository.

[3] European Commission, *Environmental Footprint (EF) 3.1 reference package*, Joint Research Centre.

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[10] Ellen MacArthur Foundation and Granta Design, *Circularity Indicators: An Approach to Measuring Circularity — Methodology*, 2015. The MCI in this report follows this formulation, including $F(X) = 0.9/X$.

[11] Regulation (EU) 2024/1781 establishing a framework for the setting of ecodesign requirements for sustainable products (ESPR).

[12] EN 45554:2020, General methods for the assessment of the ability to repair, reuse and upgrade energy-related products.

[13] European Commission, Joint Research Centre, *Product Environmental Footprint Category Rules (PEFCR) Guidance*, which defines the Circular Footprint Formula. The primary document could not be fetched directly during this work (access was blocked); the formula implemented here was reconstructed from secondary summaries of the guidance (GreenDelta, 2020; ARECO) and should be checked against the primary JRC document before being relied on beyond this report.

[14] E. Pauer, B. Wohner, M. Tacker, "The Influence of Database Selection on Environmental Impact Results. Life Cycle Assessment of Packaging Using GaBi, Ecoinvent 3.6, and the Environmental Footprint Database," *Sustainability*, 12(23), 9948, 2020. doi:10.3390/su12239948.

[15] M. Kalverkamp, E. Helmers, A. Pehlken, "Impacts of life cycle inventory databases on life cycle assessments: A review by means of a drivetrain case study," *Journal of Cleaner Production*, 269, 121329, 2020. doi:10.1016/j.jclepro.2020.121329.

[16] N. Srinivas and K. Deb, "Multiobjective Optimization Using Nondominated Sorting in Genetic Algorithms," *Evolutionary Computation*, 2(3), 221–248, 1994. The original NSGA, the direct ancestor of NSGA-II and NSGA-III.

[17] K. Deb and R.B. Agrawal, "Simulated Binary Crossover for Continuous Search Space," *Complex Systems*, 9, 115–148, 1995. The crossover operator used on the continuous genes throughout this work.

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[19] H. Jain and K. Deb, "An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point-Based Nondominated Sorting Approach, Part II: Handling Constraints and Extending to an Adaptive Approach," *IEEE Trans. Evolutionary Computation*, 18(4), 602–622, 2014. Extends NSGA-III to constrained problems, which is the form implemented here.

[20] E. Zitzler and L. Thiele, "Multiobjective Optimization Using Evolutionary Algorithms — A Comparative Case Study," in *Parallel Problem Solving from Nature — PPSN V*, Springer, 292–301, 1998. Introduces the hypervolume indicator used throughout Section 8 to compare NSGA-III, NSGA-II and random search.

[21] J. Derrac, S. García, D. Molina, F. Herrera, "A Practical Tutorial on the Use of Nonparametric Statistical Tests as a Methodology for Comparing Evolutionary and Swarm Intelligence Algorithms," *Swarm and Evolutionary Computation*, 1, 3–18, 2011. Basis for the Mann–Whitney protocol and the emphasis on effect size over p-value with small seed counts.

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[24] J. Kirchherr, D. Reike, M. Hekkert, "Conceptualizing the Circular Economy: An Analysis of 114 Definitions," *Resources, Conservation and Recycling*, 127, 221–232, 2017. Notes that most circular-economy definitions barely connect to social equity, which is part of the motivation for treating People as a full, independent objective rather than a constraint.

[25] D. Figueirinhas, Y. Vakulenko, H. Pålsson, D. Hellström, "Advancing Circularity Metrics: Revisiting the Ellen MacArthur Foundation's Material Circularity Indicator," *Resources, Conservation and Recycling*, 226, 108682, 2025. Documents specific methodological weaknesses in the MCI formula this report also implements — most relevantly its single-cycle view and its treatment of the 50:50 waste-allocation split — and is why the MCI values reported here are labelled against the published formula rather than presented as a fully resolved circularity measure.

[26] United Nations Environment Programme, *Guidelines for Social Life Cycle Assessment of Products and Organisations 2020*, UNEP/SETAC Life Cycle Initiative, 2020. The methodology the People index explicitly does not implement; cited so the distinction in Section 10 is checkable.

[27] Directive 2012/19/EU of the European Parliament and of the Council of 4 July 2012 on waste electrical and electronic equipment (WEEE) (recast). Basis for treating end-of-life collection and recovery as a first-class design variable rather than an afterthought.

[28] European Commission, *Critical Raw Materials Act and the 2023 List of Critical Raw Materials*, 2023. Aluminium's dependence on imported bauxite and the EU's low self-sufficiency ratio are part of the case for using commodity-price volatility as an observable proxy for the supply-risk term in the People index.

[29] EN 15804:2012+A2:2019, *Sustainability of construction works — Environmental product declarations — Core rules for the product category of construction products*, CEN, 2019. The closest existing standardised reporting format to what a Digital Product Passport under ESPR is expected to require; cited for context, not applied directly, since the GARO LS4 is an electrical product, not a construction product.